DEFINITION 11.3

The **least-squares estimates of slope and intercept** are obtained as follows:

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} \quad \text{and} \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

where

$$S_{xy} = \sum_i (x_i - \bar{x})(y_i - \bar{y}) \quad \text{and} \quad S_{xx} = \sum_i (x_i - \bar{x})^2$$

Thus, S_{xy} is the sum of x deviations times y deviations and S_{xx} is the sum of x deviations squared.

For the road resurfacing data, $n = 5$ and

$$\sum x_i = 1.0 + \cdots + 7.0 = 20.0$$

so $\bar{x} = \frac{20.0}{5} = 4.0$. Similarly,

$$\sum y_i = 70.0, \bar{y} = \frac{70.0}{5} = 14.0$$

Also,

$$\begin{aligned} S_{xx} &= \sum (x_i - \bar{x})^2 \\ &= (1.0 - 4.0)^2 + \cdots + (7.0 - 4.0)^2 \\ &= 20.00 \end{aligned}$$

and

$$\begin{aligned} S_{xy} &= \sum (x_i - \bar{x})(y_i - \bar{y}) \\ &= (1.0 - 4.0)(6.0 - 14.0) + \cdots + (7.0 - 4.0)(26.0 - 14.0) \\ &= 60.0 \end{aligned}$$

Thus,

$$\hat{\beta}_1 = \frac{60.0}{20.0} = 3.0 \quad \text{and} \quad \hat{\beta}_0 = 14.0 - (3.0)(4.0) = 2.0$$

From the value $\hat{\beta}_1 = 3$, we can conclude that the estimated average increase in cost for each additional mile is \$3,000.

EXAMPLE 11.2

Data from a sample of 10 pharmacies are used to examine the relation between prescription sales volume and the percentage of prescription ingredients purchased directly from the supplier. The sample data are shown here:

Pharmacy	Sales Volume, y (in \$1,000)	% of Ingredients Purchased Directly, x
1	25	10
2	55	18
3	50	25
4	75	40
5	110	50
6	138	63
7	90	42
8	60	30
9	10	5
10	100	55

a. Find the least-squares estimates for the regression line $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$.
b. Predict sales volume for a pharmacy that purchases 15% of its prescription ingredients directly from the supplier.
c. Plot the (x, y) data and the prediction equation $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$.
d. Interpret the value of $\hat{\beta}_1$ in the context of the problem.

Solution

a. The equation can be calculated by virtually any statistical computer package; for example, here is abbreviated Minitab output:

```
MTB > Regress 'Sales' on 1 variable 'Directly'

The regression equation is
Sales = 4.70 + 1.97 Directly

Predictor     Coef     Stdev    t-ratio       p
Constant     4.698     5.952       0.79   0.453
Directly    1.9705    0.1545      12.75   0.000
```

To see how the computer does the calculations, you can obtain the least-squares estimates from the following table:

	y	x	$y - \bar{y}$	$x - \bar{x}$	$(x - \bar{x})(y - \bar{y})$	$(x - \bar{x})^2$
	25	10	−46.3	−22.8	1,101.94	566.44
	55	18	−16.3	−15.8	257.54	249.64
	50	25	−21.3	−8.8	187.44	77.44
	75	40	3.7	6.2	22.94	38.44
	110	50	38.7	16.2	626.94	262.44
	138	63	66.7	29.2	1,947.64	852.64
	90	42	18.7	8.2	153.34	67.24
	60	30	−11.3	−3.8	42.94	14.44
	10	5	−61.3	−28.8	1,765.44	829.44
	100	55	28.7	21.2	608.44	449.44
Totals	713	338	0	0	6,714.60	3,407.60
Means	71.3	33.8				

$$S_{xx} = \sum (x - \bar{x})^2 = 3{,}407.6$$

$$S_{xy} = \sum (x - \bar{x})(y - \bar{y}) = 6{,}714.6$$

Substituting into the formulas for $\hat{\beta}_0$ and $\hat{\beta}_1$,

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{6{,}714.6}{3{,}407.6} = 1.9704778 \qquad \text{rounded to } 1.97$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x} = 71.3 - 1.9704778(33.8) = 4.6978519 \qquad \text{rounded to } 4.70$$

b. When $x = 15\%$, the predicted sales volume is $\hat{y} = 4.70 + 1.97(15) = 34.25$ (that is, \$34,250).

c. The (x, y) data and prediction equation are shown in Figure 11.10.

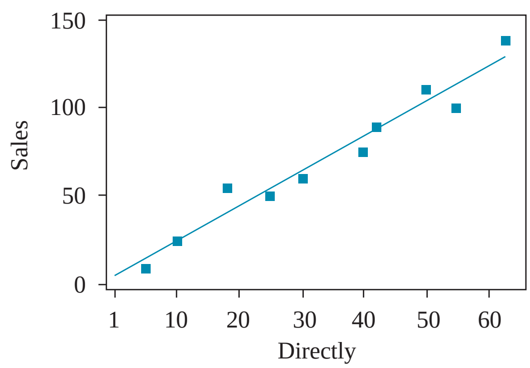

FIGURE 11.10 Sample data and least-squares prediction equation

d. From $\hat{\beta}_1 = 1.97$, we conclude that if a pharmacy would increase by 1% the percentage of ingredients purchased directly, then the estimated increase in average sales volume would be \$1,970.

EXAMPLE 11.3

Use the following Statistix output to identify the least-squares estimates for the road resurfacing data:

```
PREDICTOR
VARIABLES   COEFFICIENT   STD ERROR   STUDENT'S T   P

CONSTANT       2.00000      3.82970       0.52    0.6376
MILES          3.00000      0.85634       3.50    0.0394

R-SQUARED              0.8036   RESID. MEAN SQUARE (MSE)   14.6666
ADJUSTED R-SQUARED     0.7381   STANDARD DEVIATION         3.82970

SOURCE        DF     SS        MS        F        P

REGRESSION     1   180.000   180.000   12.27   0.0394
RESIDUAL       3   44.0000   14.6666
TOTAL          4   224.000
```

Solution The intercept is shown in the COEFFICIENT column as $\hat{\beta}_0 = 2.00000$. The slope (coefficient of x = miles) is $\hat{\beta}_1 = 3.00000$.